One Projected-Gradient Descent Iteration for the Six-Bit, Two-Check Example

Abstract

This document lists every scalar used in one projected-gradient descent iteration for the six-bit example in `p2`. The calculation transforms the six input log-likelihood ratios (LLRs) into an initial soft estimate, evaluates the penalty function corresponding to the two parity-check constraints, forms the complete gradient vector of this parity-only objective, takes a gradient step, and projects the search point onto the box constraint.

Scope. This exact projected-gradient teaching control, retained in `p2`, uses the one-soft-estimate-per-coded-bit representation of the project-defined CORAL receiver. Its objective includes only the penalty terms for the parity-check constraints: the part based on the received samples and the penalty term for the binary constraint are omitted. The step size $\eta = 0.25$ is illustrative. Production CORAL also replaces the leave-one product on a violated check by its mapped value when forming the search direction. It is therefore not simply this calculation with additional objective terms.

1 Graph, mapping, and initialization

Bits are indexed from zero. The parity-check matrix and graph neighbourhoods are

$$H = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}, \qquad \mathcal{N}(c_0) = \{1, 2, 5\}, \qquad \mathcal{N}(c_1) = \{0, 3, 4\}. \tag{1}$$

For a variable node, $\mathcal{N}(i) = \{c : i \in \mathcal{N}(c)\}$ denotes its incident checks. The source convention maps bit one to $+1$ and defines its supplied input LLR by

$$x_i = 2b_i - 1, \qquad L_i = \log \frac{P(b_i = 1 \mid y_i)}{P(b_i = 0 \mid y_i)}, \qquad x_i^{(0)} = \tanh\left(\frac{L_i}{2}\right). \tag{2}$$

Thus a positive input LLR gives a positive soft bit estimate. The hard-decision estimate at any iteration is

$$\widehat{b}_i^{(k)} = \mathbb{1}\left[x_i^{(k)} \geq 0\right]. \tag{3}$$

Because each row of (1) contains three bits, a word that satisfies the corresponding parity-check equation has bipolar product -1.

Table 1: Observations, input LLRs, and initial soft bit estimates.

i	y_i	L_i	$L_i/2$	$x_i^{(0)} = \tanh(L_i/2)$	$\widehat{b}_i^{(0)}$
0	−0.24	−0.38	−0.190	−0.187 746 206	0
1	+0.30	+0.48	+0.240	+0.235 495 750	1
2	+0.09	+0.14	+0.070	+0.069 885 890	1
3	+3.00	+4.78	+2.390	+0.983 347 814	1
4	+1.79	+2.85	+1.425	+0.890 637 366	1
5	+0.69	+1.10	+0.550	+0.500 520 211	1

The y_i values identify the original observations. This parity-only calculation takes the supplied L_i values directly and does not derive them from y_i.

The displayed transmitted word is

$$\boldsymbol{b}_{\mathrm{tx}} = (0, 1, 0, 1, 1, 1). \tag{4}$$

The initial point and its hard-decision estimate are therefore

$$\begin{aligned} \boldsymbol{x}^{(0)} = (&-0.187746206,\ 0.235495750,\ 0.069885890,\\ &0.983347814,\ 0.890637366,\ 0.500520211), \end{aligned} \tag{5}$$

$$\widehat{\boldsymbol{b}}^{(0)} = (0, 1, 1, 1, 1, 1), \qquad H\widehat{\boldsymbol{b}}^{(0)} \bmod 2 = (1, 0). \tag{6}$$

Relative to $\boldsymbol{b}_{\mathrm{tx}}$, only bit 2 is initially wrong, and check c_0 is violated.

2 Objective and one-iteration rule

For check c, define

$$p_c(\boldsymbol{x}) = \prod_{j \in \mathcal{N}(c)} x_j, \qquad r_c(\boldsymbol{x}) = 1 + p_c(\boldsymbol{x}), \qquad J_c(\boldsymbol{x}) = r_c(\boldsymbol{x})^2. \tag{7}$$

The plus sign is specific to the bit mapping and the degree-three checks in this example. For a degree-four check with the same mapping, a satisfying word has product $+1$, giving the corresponding squared residual $(1 - p_c)^2$. The objective in this calculation is

$$J_p(\boldsymbol{x}) = J_{c_0}(\boldsymbol{x}) + J_{c_1}(\boldsymbol{x}). \tag{8}$$

For $i \in \mathcal{N}(c)$, the partial derivative of J_c is

$$\frac{\partial J_c}{\partial x_i} = 2r_c(\boldsymbol{x}) \prod_{j \in \mathcal{N}(c) \setminus \{i\}} x_j. \tag{9}$$

The gradient of the objective function is obtained by adding the partial derivatives from the checks that contain bit i:

$$g_i^{(0)} = \left.\frac{\partial J_p}{\partial x_i}\right|_{\boldsymbol{x}^{(0)}} = \sum_{c \in \mathcal{N}(i)} \left.\frac{\partial J_c}{\partial x_i}\right|_{\boldsymbol{x}^{(0)}}. \tag{10}$$

All six partial derivatives are evaluated at the same $\boldsymbol{x}^{(0)}$ before any variable is changed. The Tanner graph specifies which variables enter each product; these derivatives are not directed variable-to-check or check-to-variable messages, and they are not retained as state on directed edges.

One projected-gradient descent iteration consists of the gradient step and the projection step

$$z_i^{(0)} = x_i^{(0)} - \eta g_i^{(0)}, \qquad \eta = 0.25, \tag{11}$$

$$x_i^{(1)} = \Pi_{[-1,1]}\left(z_i^{(0)}\right) = \min\left(1, \max(-1, z_i^{(0)})\right). \tag{12}$$

The Euclidean projection in (12) enforces only the box constraint $\boldsymbol{x} \in [-1, 1]^6$; it does not enforce parity.

3 Objective at the initial point

Table 2 evaluates every quantity used by (9). The factor $2r_c$ is included because it is shared by the three partial derivatives associated with check c.

Table 2: Initial check products, residuals, and penalty terms.

check	bits	$p_c^{(0)}$	$r_c^{(0)}$	$2r_c^{(0)}$	$J_c^{(0)}$	parity status
c_0	1, 2, 5	+0.008 237 477	1.008 237 477	2.016 474 953	1.016 542 809	violated
c_1	0, 3, 4	−0.164 429 311	0.835 570 689	1.671 141 378	0.698 178 376	satisfied

Hence

$$J_p(\boldsymbol{x}^{(0)}) = 1.016542809238 + 0.698178376170 \approx 1.714721185407. \tag{13}$$

4 All partial derivatives at the initial point

Define the product that excludes variable i from check c as

$$q_{c,i}^{(0)} = \prod_{j \in \mathcal{N}(c) \setminus \{i\}} x_j^{(0)}. \tag{14}$$

Every variable belongs to one check in this example, so $g_i^{(0)} = \partial J_c / \partial x_i = 2 r_c^{(0)} q_{c,i}^{(0)}$.

Table 3: All products and partial derivatives used to form the initial gradient vector.

i	check	other bits	$q_{c,i}^{(0)}$	$2r_c^{(0)}$	$g_i^{(0)}$	$-g_i^{(0)}$
0	c_1	3, 4	+0.875 806 306	1.671 141 378	+1.463 596 157	−1.463 596 157
1	c_0	2, 5	+0.034 979 301	2.016 474 953	+0.070 534 884	−0.070 534 884
2	c_0	1, 5	+0.117 870 382	2.016 474 953	+0.237 682 674	−0.237 682 674
3	c_1	0, 4	−0.167 213 786	1.671 141 378	−0.279 437 877	+0.279 437 877
4	c_1	0, 3	−0.184 619 821	1.671 141 378	−0.308 525 822	+0.308 525 822
5	c_0	1, 2	+0.016 457 830	2.016 474 953	+0.033 186 802	−0.033 186 802

Thus

$$\begin{aligned} \nabla J_p(\boldsymbol{x}^{(0)}) = (&1.463596157,\ 0.070534884,\ 0.237682674,\\ &-0.279437877,\ -0.308525822,\ 0.033186802). \end{aligned} \tag{15}$$

5 Gradient step and projection step

The change applied by the gradient step is $\Delta x_i^{(0)} = -\eta g_i^{(0)}$. Table 4 lists every multiplication and addition in (11).

Table 4: Gradient step with $\eta = 0.25$.

i	$x_i^{(0)}$	$g_i^{(0)}$	$\Delta x_i^{(0)} = -0.25 g_i^{(0)}$	$z_i^{(0)} = x_i^{(0)} + \Delta x_i^{(0)}$
0	−0.187 746 206	+1.463 596 157	−0.365 899 039	−0.553 645 245
1	+0.235 495 750	+0.070 534 884	−0.017 633 721	+0.217 862 029
2	+0.069 885 890	+0.237 682 674	−0.059 420 668	+0.010 465 222
3	+0.983 347 814	−0.279 437 877	+0.069 859 469	+1.053 207 283
4	+0.890 637 366	−0.308 525 822	+0.077 131 456	+0.967 768 821
5	+0.500 520 211	+0.033 186 802	−0.008 296 701	+0.492 223 511

Table 5 then applies the projection operator. Only coordinate 3 lies outside the box.

Table 5: Projection onto $[-1, 1]^6$ and resulting hard-decision estimates.

i	$z_i^{(0)}$	$x_i^{(1)}$	$x_i^{(1)} - z_i^{(0)}$	clipped?	$\widehat{b}_i^{(1)}$
0	−0.553 645 245	−0.553 645 245	+0.000 000 000	no	0
1	+0.217 862 029	+0.217 862 029	+0.000 000 000	no	1
2	+0.010 465 222	+0.010 465 222	+0.000 000 000	no	1
3	+1.053 207 283	+1.000 000 000	−0.053 207 283	yes	1
4	+0.967 768 821	+0.967 768 821	+0.000 000 000	no	1
5	+0.492 223 511	+0.492 223 511	+0.000 000 000	no	1

The point after one complete projected-gradient descent iteration is

$$\begin{aligned} \boldsymbol{x}^{(1)} = (&-0.553645245,\ 0.217862029,\ 0.010465222,\\ &1.000000000,\ 0.967768821,\ 0.492223511). \end{aligned} \tag{16}$$

6 State after one iteration

Re-evaluating the two penalty terms at $\boldsymbol{x}^{(1)}$ gives Table 6.

Table 6: Check values after the first projected-gradient descent iteration.

check	bits	$p_c^{(1)}$	$r_c^{(1)}$	$2r_c^{(1)}$	$J_c^{(1)}$	parity status
c_0	$1, 2, 5$	$+0.001\,122\,257$	$1.001\,122\,257$	$2.002\,244\,514$	$1.002\,245\,774$	violated
c_1	$0, 3, 4$	$-0.535\,800\,606$	$0.464\,199\,394$	$0.928\,398\,787$	$0.215\,481\,077$	satisfied

Therefore

$$J_p(\boldsymbol{x}^{(1)}) = 1.217726850659, \tag{17}$$

$$J_p(\boldsymbol{x}^{(0)}) - J_p(\boldsymbol{x}^{(1)}) = 0.496994334749 \quad (28.984\% \text{ relative to } J_p(\boldsymbol{x}^{(0)})). \tag{18}$$

The objective value decreases, but $x_2^{(1)} = +0.010465222$ remains on the positive side. Consequently,

$$\widehat{\boldsymbol{b}}^{(1)} = (0, 1, 1, 1, 1, 1), \qquad H\widehat{\boldsymbol{b}}^{(1)} \bmod 2 = (1, 0). \tag{19}$$

The hard-decision estimate and syndrome are unchanged: one iteration does not yet correct bit 2. In contrast, one sum-product message-passing iteration on this same example changes bit 2 and recovers $\boldsymbol{b}_{\text{tx}}$.

Each variable node in this small graph has degree one, so the graph is a pair of disconnected check stars. It cannot demonstrate the aggregation of several incident-check contributions into one gradient coordinate, and any numerical similarity to sum-product does not establish equivalence.

Recomputed partial derivatives at $\boldsymbol{x}^{(1)}$

These values describe the state available to the next iteration; they are not used retroactively in the first update.

Table 7: Products and partial derivatives recomputed after the first iteration.

i	check	other bits	$q_{c,i}^{(1)}$	$2r_c^{(1)}$	$g_i^{(1)}$	$-g_i^{(1)}$
0	c_1	$3, 4$	$+0.967\,768\,821$	$0.928\,398\,787$	$+0.898\,475\,400$	$-0.898\,475\,400$
1	c_0	$2, 5$	$+0.005\,151\,228$	$2.002\,244\,514$	$+0.010\,314\,019$	$-0.010\,314\,019$
2	c_0	$1, 5$	$+0.107\,236\,813$	$2.002\,244\,514$	$+0.214\,714\,320$	$-0.214\,714\,320$
3	c_1	$0, 4$	$-0.535\,800\,606$	$0.928\,398\,787$	$-0.497\,436\,633$	$+0.497\,436\,633$
4	c_1	$0, 3$	$-0.553\,645\,245$	$0.928\,398\,787$	$-0.514\,003\,574$	$+0.514\,003\,574$
5	c_0	$1, 2$	$+0.002\,279\,974$	$2.002\,244\,514$	$+0.004\,565\,066$	$-0.004\,565\,066$

Thus the gradient ready for iteration two is

$$\begin{aligned}\nabla J_p(\boldsymbol{x}^{(1)}) = (&0.898475400,\ 0.010314019,\ 0.214714320,\\ &-0.497436633,\ -0.514003574,\ 0.004565066).\end{aligned} \tag{20}$$

Numerical provenance. Within `JadeJcade`, the graph, observations, LLRs, and transmitted word come from `paper/p2/p2_assets/frames/os00_exact_example.tex`. The objective and illustrative step come from `paper/p2/p2_assets/frames/os00b_projected_gradient.tex`; the retained executable checker, `exact_six_bit_example.py`, is under `paper/one_scalar_per_bit_slides/one_scalar_per_bit_slides_assets/examples/`. Every displayed value was independently recomputed from the rounded input LLRs. A symmetric numerical derivative check with perturbation 10^{-6} agrees with $\nabla J_p(\boldsymbol{x}^{(0)})$ to within 1.92×10^{-10} in every coordinate.